

CS 4412 - M3 Analysis Document

Project: Sleep Health and Lifestyle Pattern Discovery

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This document summarizes the complete M3 implementation, analytical choices, current findings, discovery-question progress, and the plan for M4 polish.

1. Implementation Status

The M3 pipeline is now working end to end. The analysis loads the real-world sleep and lifestyle dataset, cleans the fields, handles missing sleep-disorder values, splits blood pressure into systolic and diastolic measures, standardizes numeric variables, and runs multiple mining techniques. The implemented techniques are clustering comparison, PCA, interpretable classification, association rules, and anomaly detection.

2. Analytical Choices

The project moved beyond the simulated M2 fatigue dataset and switched to the Sleep Health and Lifestyle dataset so that the M3 results would rest on real data rather than artificial structure. For clustering, K-Means and Agglomerative were compared directly, while DBSCAN was included to test density-based structure. K-Means was tuned over $k = 2$ through 6 using silhouette scores, with the best K-Means score occurring at $k = 6$. PCA was chosen for low-dimensional visualization. Decision Tree and Naive Bayes were used as discovery-support tools to explain cluster structure rather than as deployment models. Association rules were mined from discretized variables to answer co-occurrence questions, and LOF was used for anomaly detection because it aligns well with the course emphasis on unusual profiles.

3. Results

The clustering comparison found the following silhouette scores: DBSCAN=0.886, Agglomerative=0.540, KMeans=0.537. The numeric K-Means profiles show that some clusters are marked by shorter sleep, higher stress, and poorer sleep quality, while others show longer sleep, lower stress, and better sleep quality. The strongest association rules highlight recurring connections among occupation, sleep-disorder status, sleep-duration buckets, stress level, and BMI group. LOF flagged 18 anomalies, which can be investigated as unusual sleep-health cases. The decision tree and Naive Bayes models both aligned strongly with the discovered K-Means labels, which suggests that the cluster structure is reflected clearly in the numeric variables used for discovery.

4. Discovery Question Progress

The original proposal asked about natural groupings, interpretable conditions associated with outcomes, and unusual response profiles. M3 now addresses those questions more directly. Natural groupings are represented by the cluster profiles and PCA projections. Conditions that tend to appear together are represented by the association-rule output. Unusual cases are represented by the LOF anomaly list. Compared with M2, the project now offers stronger evidence because the data is real-world rather than simulated, which makes the patterns more meaningful for interpretation.

5. Interpretation

The current results suggest that sleep quality is not driven by one factor alone. Instead, sleep duration, stress, activity, cardiovascular measures, and disorder labels interact in recognizable combinations. In

domain terms, some groups appear to reflect healthier and more balanced routines, while others reflect more strained profiles with shorter sleep and higher stress. The anomalies matter because they may identify edge cases such as people whose sleep-related outcomes differ from what the surrounding data would suggest.

6. M4 Plan

M4 should focus on polish, critical assessment, and presentation. The notebook already works, so the next step is to tighten the visual story: select the clearest cluster plots, reduce any redundant outputs, highlight the most meaningful association rules, and sharpen the written interpretation. The final report should also discuss limitations such as dataset size, possible repeated occupation patterns, and the difference between descriptive discovery and causal claims.

Selected M3 Metrics

Metric	Value
Rows in M3 dataset	374
K-Means best k	6
K-Means silhouette	0.537
Agglomerative silhouette	0.540
Decision Tree accuracy	1.000
Naive Bayes accuracy	1.000
LOF anomalies	18